Quiz: Instruction formats

This quiz has 5 questions. The passing grade is 4. Back Next

Started on	Wednesday, January 15, 2025, 12:12 AM
State	Finished
Completed on	Wednesday, January 15, 2025, 12:15 AM
Time taken	3 mins 29 secs
Grade	**2.00** out of 5.00 (**40**%)

Question **1**

Incorrect

0.00 points out of 1.00

Instruction operands can be (select the three correct options)

Select one or more:

- [x] A. Memory addresses ✔
- [ ] B. Immediate
- [x] C. Registers ✔
- [ ] D. The PSW
- [x] E. Operation codes ✖

Instruction operands are:

- Registers in RR-, RX-, RXY-, and RI-type instructions
- Storage addresses in RS-, RX-, RXY-, SI-, and SS-type instructions
- Immediate values in SI- and RI-type instructions

The PSW is a register, not an operand in an instruction.

Operation codes, as the name indicates, are the operation (ADD, LOAD, MULTIPLY...) to perform, not an operand.

The correct answers are: Registers, Memory addresses, Immediate

Question **2**

Correct

1.00 points out of 1.00

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z/Architecture instructions are 1, 2, or 3 halfwords in length, and always start on a halfword boundary

Select one:

- True ✔
- False

Correct!

The correct answer is 'True'.

Question **3**

Incorrect

0.00 points out of 1.00

In an RS-format instruction

Select one:

- A. One operand is a register and another a long displacement storage address
- B. Both operands are registers ✖
- C. One operand is a register and another a base-displacement storage address
- D. One operand is a register and another an indexed storage address
- E. One operand is a storage address and another an immediate value
- F. One operand is a register and another a 16-bit relative address
- G. Both operands are storage addresses

The correct answer is: One operand is a register and another a base-displacement storage address

Question **4**

Incorrect

0.00 points out of 1.00

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If the index of an RX-type second operand is register zero, the effective address consists of: base register contents + register zero (index) contents + displacement

Select one:

- True ✖
- False

If the index register of an RX-type second operand is zero, the effective address consists of:

base register contents + **0 (index register ignored)** + displacement

If the base register is also zero, the effective address is just the displacement:

0 (base register ignored) + **0 (index register ignored)** + displacement

The correct answer is 'False'.

Question 5

Correct

1.00 points out of 1.00

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The differences between RXY-type instructions and RX-type instructions are:

1. RXY-type instructions have long-displacement (20-bit) second operands instead of base-displacement (12-bit) second operands
2. RXY-type instructions have an extended (16 bit) operation code

Select one:

- [x] True ✔
- [ ] False

Correct!

The correct answer is 'True'.